

2.3 Nature of quantities

This section provides knowledge and understanding of scalars and vectors quantities. Vector quantities add and subtract very differently to scalar quantities; hence

it is important to know whether a quantity is a vector or a scalar.

2.3.1 Scalars and vectors

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) scalar and vector quantities	Learners will also be expected to give examples of each.
(b) vector addition and subtraction	
(c) vector triangle to determine the resultant of any two coplanar vectors	To be done by calculation or by scale drawing <i>M0.6, M4.2, M4.4</i>
(d) resolving a vector into two perpendicular components; $F_x = F \cos \theta$; $F_y = F \sin \theta$.	<i>M0.6, M4.5</i>